Total Marks of Que | Examiner

VIT
Autonomous College Affiliated to University of Mumbai

End Semester Examination (R-2022 Scheme) -(2024-25)

Date 19 / 11 / 2024	Branch CMPN / INFT	Time 1 Hr & 45 Mints
Semester 5	Subject Software Engineering / Software Engineering with WDL	Marks 40
N.B. :- All Questions are Compulsory		CO

Q. 1)	Attempt any two (5 Marks Each)	
a)	How does XP address challenges in coding, testing, and collaboration within Agile frameworks?	CO 1
b)	How does requirement engineering help in mitigating project risks? Explain with one example.	
c)	Explain W5HH principles.	CO 5
Q. 2)	Attempt any One (10 Marks Each)	
a)	Explain Management Spectrum with 4P's of Software project management.	CO 3
b)	Create a website registration form with Name ,address , Phone no , email id with HTML and JavaScript for validating user input to ensure fields meet specified requirements.	CO 6
Q 3)	Attempt any One (10 Marks Each)	

a)	#include <stdio.h>
	main()
	{
	float x,y,z;
	clrscr();
	printf("enter the three variables x,y,z");
	scanf("%f %f %f",&x,&y,&z);
	if(x > y)
	{
	if(x > z)
	printf("x is greatest");
	else
	printf("z is greatest");
	}
	else
	{
	if(y > z)
	printf("y is greatest");
	else
	printf("z is greatest");
	}
	getch();
	}
	
	(a) Draw the DD graph for the program.
	(b) Calculate the cyclomatic complexity of the program
	(c) List all independent paths.
		CO 4

b)	Explain Cohesion and Coupling ? Explain different types of cohesion.	CO 3
Q 4)	Attempt any One (10 Marks Each)	
a)	Define Risks. What are the different categories of risks? Explain RMMM plan with suitable example.	CO 4
b)	For a project of 2000LOC of semidetached type project with highly capable programmers(0.86), with high reliability(1.15) and moderate use of software tools(1.00) Calculate efforts, time, no. of people required also Calculate cost of project based on efforts if salary of person is Rs. 15000/month a 3.0 , b 1.12 , c 2.5 , d 0.35	CO 2

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	Q 1A]	How does XP address challenges in coding, testing and collabration within agile framework.

Ans: Extreme programming (XP) takles challenges in coding, testing and collabration within Agile framework, by emphasizing practices like pair programming, test driven development, countinuous integrat°, which promote high quality code rapid iteration, and seamless communication between team members and stakeholders. This allows for quick adaption to changeing requirements throughout development process.

XP coding

- pair programming

Two developers work together on the same task, constantly reviewing each others code leading to better quality & less bugs.

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Simplicity

Focus on creating clean, easy to understand code, avoiding unnecessary complexity.

Refactoring

Countinuosly improving code structure without changing its functionality ensuring maintainability.

XP Testing

Test Driven Development TDD:

writing automated tests before writing code, ensuring features are well defined and tested.

Countinuous Integration

Regularly integrating code changes into the main code base, quickly identifying integration issues.

Customer Involvement

Having customer actively participate in development process, providing frequent feedback on features & priorites.

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	Q1B]	How does requirement engineering help in mitigating project risks? Explain with one example.

Ans: Requirement engineering helps mitigate project risk by clearly defining and documenting the projects needs early on, allowing for proactive identification and addressing of potential issues before development begins. This can prevent costly changes later in the project lifecycle.

Requirement engineering involves systematically gathering, analyzing, documenting & validating project requirements

It mitigates risk by:

1. Clarifying Ambiguities
   ensures all stakeholders have a common understanding of project objectives.

2. prioritizing Requirements.
   focus on critical needs, reducing resource wastage.

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3. Managing scope:
prevent scope creep by defining clear boundaries.

4. Reducing Rework:
Easly detecting of potential issues avoids costly revisions

Example

In a healthcare software project requirement engineering identifies madatory features like patient data security and compliance with regulations.

clearing documenting these ~~documents~~ reduces legal risks and prevents future compliance related work.

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Q1C] Explain W5HH principles.

Ans Barry Boehm's W5HH — Why, What, When, Who, where, How, How much. principles provide a structured approach to project management.

1. WHY
Understanding the underlying reasons for developing the s/w helps in aligning the project with overall goals & objects.

2. WHAT
What functionalities and features does the s/w need to provide? Defining deliverables helps track progress.

3. WHEN :
Understanding time constraints helps in setting realistic expectations and planning the development process accordingly.

4: WHO :
It is important to identify the people or groups that s/w will serve

and consider their specific needs, roles and responsibilities.

5. WHERE:
Where will the s/w be used? whether its web application, desktop / Mobile app.
The environment can impact design & functionality considered.

6. HOW:
It relates to technical aspect, development methodologies, tools that how will the s/w be developed.

7. HOW MUCH
This is about budget & resources required for project involving cost & resource estimations.

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Q.2A] Explain Management spectrum with 4P's of software project management.

Ans: The Management Spectrum covers various aspects like project management, team management, process management etc.

Different methodologies & approaches can be combined to create a management strategy that aligns with specific needs of project & organization.

4Ps of Management Spectrum are

- people
- product
- process
- project

people

This aspect focuses on managing the human resources involved in the project emphasizing team dynamix, leadership & stakeholder management.

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It includes activities such as defining roles and responsibilities assembling the right team with required skills, fostering a collaborative and motivating work environment, provide training & professional development opportunities & addressing any interpersonal or team related challenges.

Managing interpersonal conflict, keeping remote or diverse teams aligned, & balancing workload are the major challenges.

Product:

Managing the product Aspect involves defining, designing and delivering the software product that meets the stakeholder's requirements and expectations.

This includes task such as gathering and prioritizing require ments, creating a detailed

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design, ensuring proper coding and testing practices. and validating the final product aligns with intended functionality & quality standards.

Aligning the product with changing client requirements, Maintaining quality under tight deadlines or budget constraints, Managing client or stakeholder feedback are the major challenges under product aspect.

PROJECT

It is the foundation of project management effort. what is being done and sets boundaries within which the team will operate.

This includes planning the project scope, defining the milestones & deliverables, estimating resources and timeframes, creating a project schedule

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monitoring progress, identifying & mitigating risks and ensuring that the project stays on track & within budget.

Effective project management ensures that the project's goals are archived while managing constraints & changes.

PROCESS

The process aspect focuses on defining and optimizing the s/w development processes used throughout the project lifecycle.

This involves selecting proper methods, standards, version control setting up testing practices & ensuring that team follows best practices for efficient and high quality development.

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Q2B]

Ans

```
<DOCTYPE html>
<html>
<head>
<title> Registeration Form </title>
<script>

function validationform() {

const name = document.form["regform"]
            ["name"].value;
const address = document.form["regform"]
            ["address"].value;
const phone = document.form["regform"]
            ["phone"].value;
const email = document.form["regform"]
            ["email"].value;

const phoneRegex = /^\d{10}$/;
const emailRegex = /^[^/s@]+@[^\s@]+\
            .[^\s@]+$/;

if (name.trim() == "" || address.trim()
   == "" || phone.trim() == "" || email.
   trim() == "  ") {
alert ("All fields are required!");
   return false;
}
```

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```
if (!phoneRegex.test(phone)) {
    alert("Phone number must be
    exactly 10 digits.");
    return false;
}

if (!emailRegex.test(email)) {
    alert("Invalid email format.
    Example: user@example.com");
    return false;
}

alert("Registration Successful!");
return true;
}
</script>
</head>
<body>
    <h2> Registration Form </h2>

<form name = "regForm"
    onsubmit = "return validateForm();">

<label for = "name"> Name: </label>
    <br>

<input type = "text" id = "name"
    name = "name"> <br> <br>
```

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```
<label for = "address">Address:
    .</label><br>
<textarea id = "address" name =
"address"></textarea><br><br>

<label for = "phone" > phone No.
        </label> <br>
<input type = "text" id = "phone"
 name = "phone" maxlength = "10">
     <br><br>

<label for = "email"> Email ID:
     </label> <br><br>

<input type = "text" id = "email"
  name = "email"> <br><br>

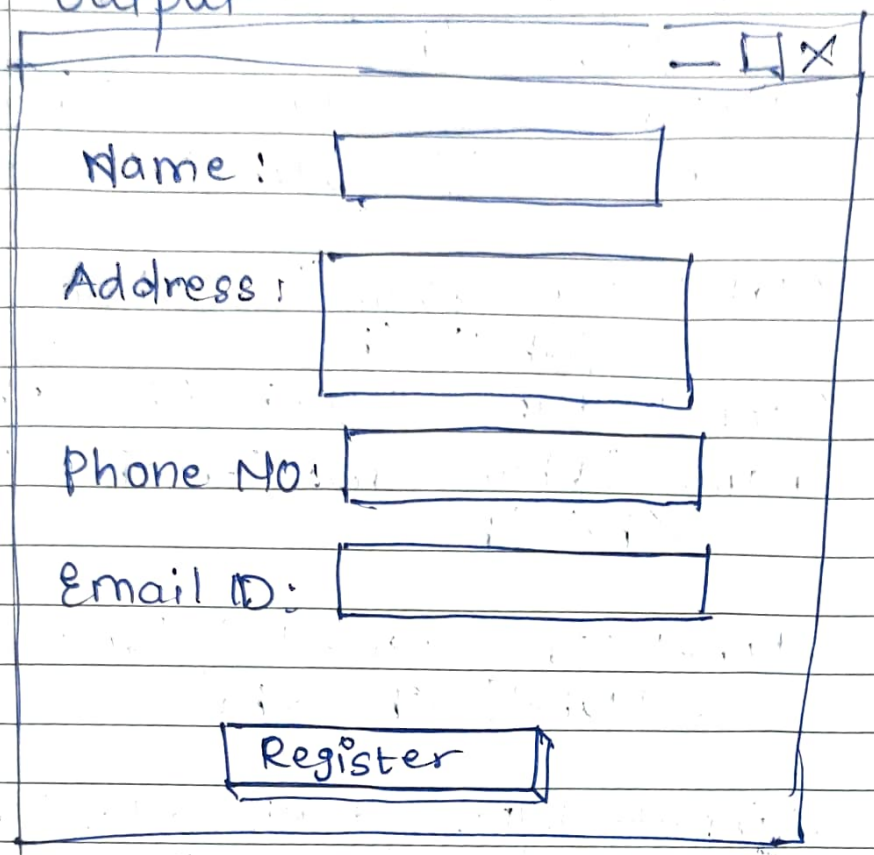
<button type = "Submit" >
  Register </button>

</form>
</body>
</html>
```

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Output :



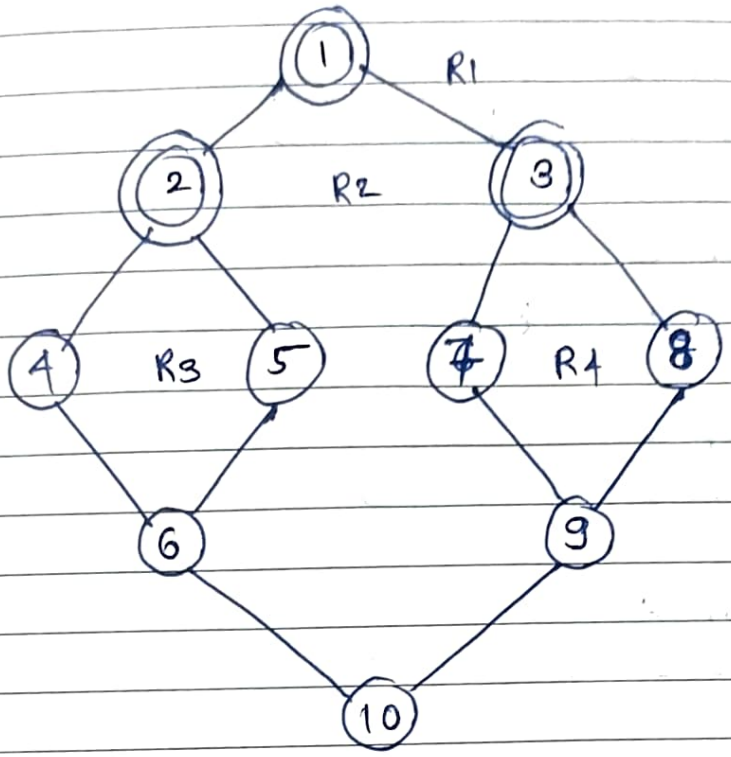
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Q3A]

(a).



b] i) cyclomatic complexity.

$V(G) = E - N + 2$

$= 12 - 10 + 2$

$= 4$ //

ii) total no. of Regions are 4 //

iii) $V(G)$ = predicate + 1

$= 3 + 1$

$= 4$

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c] Independent path are

path 1.
1 - 2 - 4 - 6 - 10

path 2
1 - 2 - 5 - 6 - 10

path 3
1 - 3 - 7 - 9 - 10

path 4
1 - 3 - 8 - 9 - 10.

Q3b] Explain cohesion and coupling Explain different types of cohesion.

Ans: Cohesion and Coupling are two critical concepts in software design that define quality of module.

Cohesion :

It refers to degree to which the elements within a module belong together.

A highly cohesive modules has elements that perform closely related tasks or work towards a single purpose.

High cohesion is desirable as it makes a module easier to understand, maintain and reuse.

A cohesive module performs a single task within a slw procedure requireing little interaction with procedures being performed in other parts of program.

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Coupling

Coupling is a measure that defines the level of inter dependability among modules of a program.

It tells at what level the modules interfere and interact with each other.

The lower the coupling, the better the program.

Coupling depends on the interface complexity between modules, the point at which entry or reference is made to a module and what data pass across interface.

Coupling is Inter module concept.

Types of Cohesion

1. Coincidental Cohesion

It is unplanned and random cohesion, which might be result of breaking the program into smaller modules for sake of modularization

E.g: A module that combines unrelated tasks like printing, database connection and file reading.

2] Logical Cohesion

Elements perform similar types of tasks but are executed based on control flag.

E.g: A module with a single function to handle all I/O operat^n based on parameter.

3] Temporal Cohesion

Elements are grouped because they are executed at same time or

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during a specific phase.

E.g: A module that initializes variables, opens log files and sets configuration parameters at start of program.

4] Procedural Cohesion

Elements are grouped because they follow a specific sequence of execution.

E.g: A module handling steps for processing a customer order.
validating input - calculating total - and generating receipts.

5] Communicational Cohesion

Elements operate on the same data or contribute to a single task.

E.g: A module that reads a customer record, update it

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saves it back to database.

6] Functional Cohesion

Elements work together to achieve a single, well defined purpose.

E.g: A module that calculates square root of number.

7] Informational Cohesion

A module that performs multiple tasks but each task operates on same data.

Eg: A module for handling employee records that allows adding, updating & deleting records.

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Q4A] Define Risk. What are the different categories of Risks.? Explain RMMM plan with suitable Example

Ans'

In slw development, A Risk is a potential problem that might occur in future and negatively impact project's success.

Risks are uncertain but can have consequense such as cost overruns, delays or reduced quality it not-managed effectively.

Categories of Risk.

- Project Risk

Risk that affect project schedule, resources are budget

- Technical Risk

Risks related to technology, tools or technical implemention.

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- **Business Risk**

Risk impacting the viability of software in the business environment.

- **Operational Risk**

Risk related to deployment, operation or maintenance of software.

- **External Risk**

Risk from external factors like regulatory changes, competions, or natural disasters etc.

An RMMM plan is a strategy to address risk by mitigating their likelihood and impact, monitoring their occurance, and managing their effects if they materialize.

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Example

Risk:

Delay in delivery due to resource unavailability.

Mitigation Plan

- Cross train team members to handle multiple tasks
- Maintain buffer in project overall schedule

Monitoring plan

- weekly check on resource availability & task progress
- Daily status tracking.

Management plan

- Reassign task to other team members
- Hire temporary staff only if risk occurs.

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	Q4B	

project type: semi detached

LOC = 2000

∴ KLOC = 2

Efforts:

$E = a \cdot (KLOC)^b \times EAF$

$= 3.0 \times 2^{1.12} \times (0.86 \times 1.15 \times 1.00)$

$= 6.84$ person month

$\cong$ 9 person month

Time:

$T = C \cdot (E)^d$

$= 2.5 (6.84)^{0.35}$

$= 3.94$

$\cong$ 4 months.

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No. of people = $\frac{\text{Effort}}{\text{Time}}$

$= \frac{6.85}{3.94}$

$= 1.74$ //

$\simeq$ 2 people.

Cost

= E x T x salary

= 9 x 15000 x 4

= 36 x 15000

= Rs. 5,40,000/-